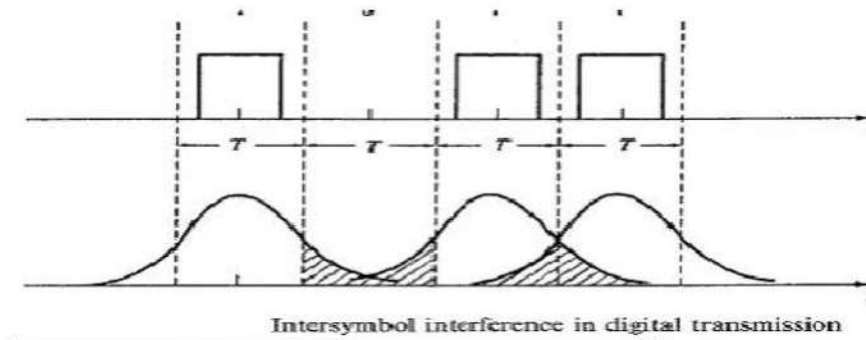


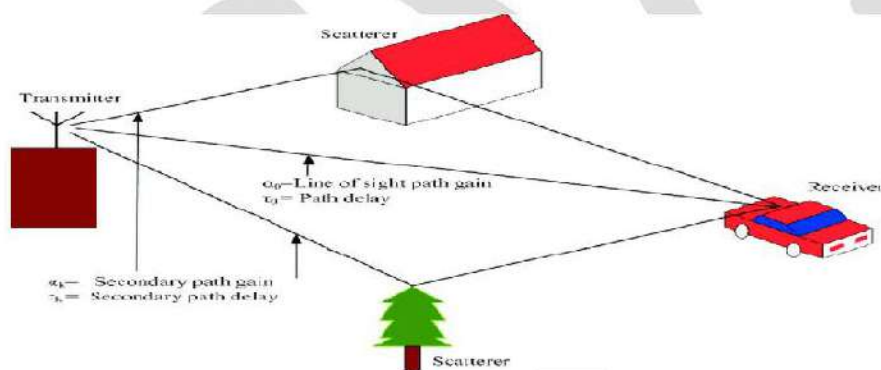
MODULE 5

INTER SYMBOL INTERFERENCE

Inter symbol interference (ISI) is when signals overlap and mix together, causing confusion in the information being sent. By managing the timing and using special techniques like spacing filtering and equalization (smoothing the wave) ISI can be reduced.



ISI is usually caused by multipath propagation or the inherent linear or non-linear frequency response of a communication channel causing successive symbols to blur together.



The source of bit errors in a base band-pulse transmission is inter symbol interference (ISI), which arises when the communication channel is dispersive. When we say a channel is dispersive, we mean the channel has a frequency dependent amplitude spectrum.

The simplest example of a dispersive channel is the band-limited channel. For example, a brick wall band-limited channel passes all frequencies $|f| < W$ without distortion, while it blocks all frequencies $|f| > W$.

For the baseband transmission of digital data, the use of discrete pulse-amplitude modulation (PAM) is the most efficient one in terms of power and bandwidth utilization.

In PAM, the amplitude of each pulse represents the data being transmitted. This makes it easy to visualize how different pulses can interfere with each other.

The pulses in PAM are distinct and their shapes can be easily observed, which helps in illustrating how they overlap and cause ISI.

MODULE - 5

Syllabus:

Baseband Transmission of Digital Signals: Introduction, Intersymbol Interference, Eye Pattern, Nyquist Criterion for distortionless Transmission, Baseband M-ary PAM Transmission.

Noise: Signal to Noise Ratio, External Noise, Internal Noise, Semiconductor Noise, Expressing Noise Levels, Noise in Cascade Stages.

*. Introduction:

Transmission requires the use of a low pass channel with a bandwidth large enough to accommodate the essential frequency content of the data stream.

The result of data transmission over such a channel is that each received pulse is affected somewhat by adjacent pulses, thereby giving rise to a common form of interference called Intersymbol Interference (ISI).

Noise and ISI arise in the system simultaneously which leads to Bit errors.

*. Intersymbol Interference (ISI)

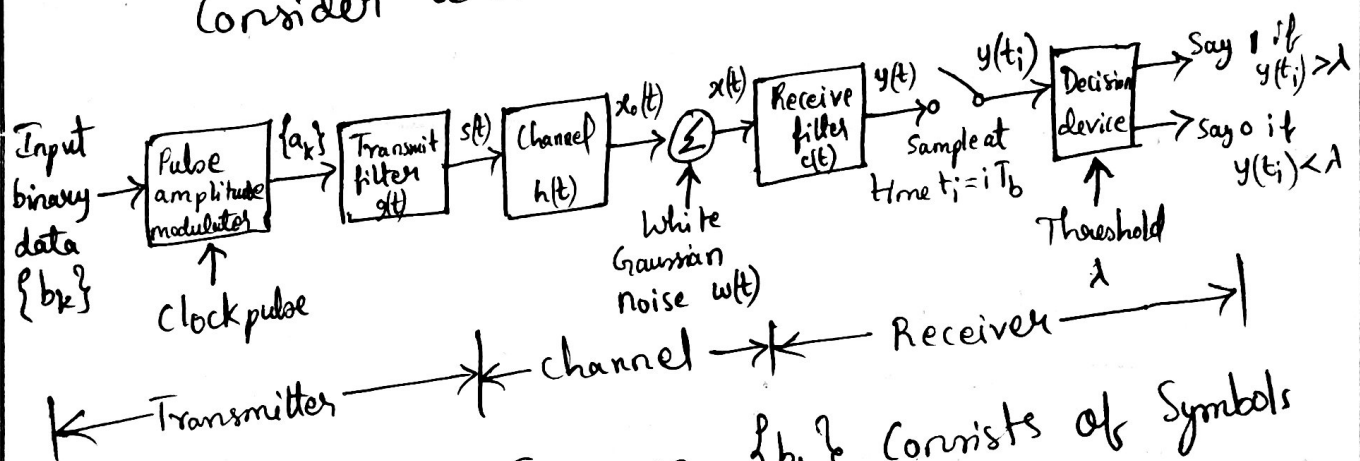
①. Define Intersymbol Interference (ISI). outline Baseband binary data transmission system with neat block diagram and equations. (8 marks)

→

ISI is a phenomenon that occurs in digital Communication Systems where Symbols (data) transmitted over a channel interfere with each other due to various factors such as channel dispersion, multipath propagation or insufficient bandwidth.

For base-band transmission of digital data, use of discrete pulse-amplitude modulation (PAM) is the most efficient one in terms of power and bandwidth utilization.

Consider a baseband binary PAM System,



The incoming binary sequence $\{b_k\}$ consists of Symbols '1' & '0' each of duration T_b .

PAM output is $\{a_k\} = \begin{cases} +1 & \text{if Symbol } b_k \text{ is 1} \\ -1 & \text{if Symbol } b_k \text{ is 0.} \end{cases}$

This Sequence of short pulses is applied to a transmit filter of impulse response $g(t)$, producing transmitted signal

$$s(t) = \sum_k a_k g(t - kT_b) \quad \text{--- (1)}$$

$S(t)$ is transmitted through the channel of impulse response $h(t)$. In addition, channel adds random noise to the signal at the receiver input.

Noisy signal $x(t)$ is then passed through a receive filter of impulse response $c(t)$. The resulting filter output $y(t)$ given by

$$y(t) = \mu \sum_k a_k p(t - kT_b) + n(t) \quad (2)$$

Where $\mu \rightarrow$ Scaling factor
 $p(t) \rightarrow$ pulse to be defined.

To be precise, the Scaled pulse $\mu p(t)$ is given by

$$\mu p(t) = g(t) * h(t) * c(t) \quad (3)$$

$* \rightarrow$ Convolution in Time Domain

Time domain Convolution is transformed into Multiplication in frequency domain.

$$\therefore \mu P(f) = G(f) \cdot H(f) \cdot C(f) \quad (4)$$

The term $n(t)$ in equation (2) is noise produced at the output of the receive filter, due to the additive noise $w(t)$.

The receive filter $y(t)$ is sampled at time $t_i = iT_b$ which leads to

$$y(t_i) = \mu \sum_{k=-\infty}^{\infty} a_k p[(i-k)T_b] + n(t_i)$$

$$y(t_i) = \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p[(i-k)T_b] + n(t_i) \quad (5)$$

the first term, ua_i represents the contribution of the i th transmitted bit.

the second term, represents the residual effect of all other transmitted bits on the decoding of the i th bit; This residual effect due to the occurrence of pulses before and after the sampling instant t_i is called intersymbol interference (ISI).

The last term, $n(t_i)$ represents the noise sample at time t_i .

In the absence of both ISI and noise

$$\boxed{y(t_i) = ua_i}$$
 which shows that

Under these ideal conditions, the i th transmitted bit is decoded correctly.

(*) EYE PATTERN

②. Define Eyepattern. Explain the Significance of (6 marks)
Eye diagram with a real diagram.

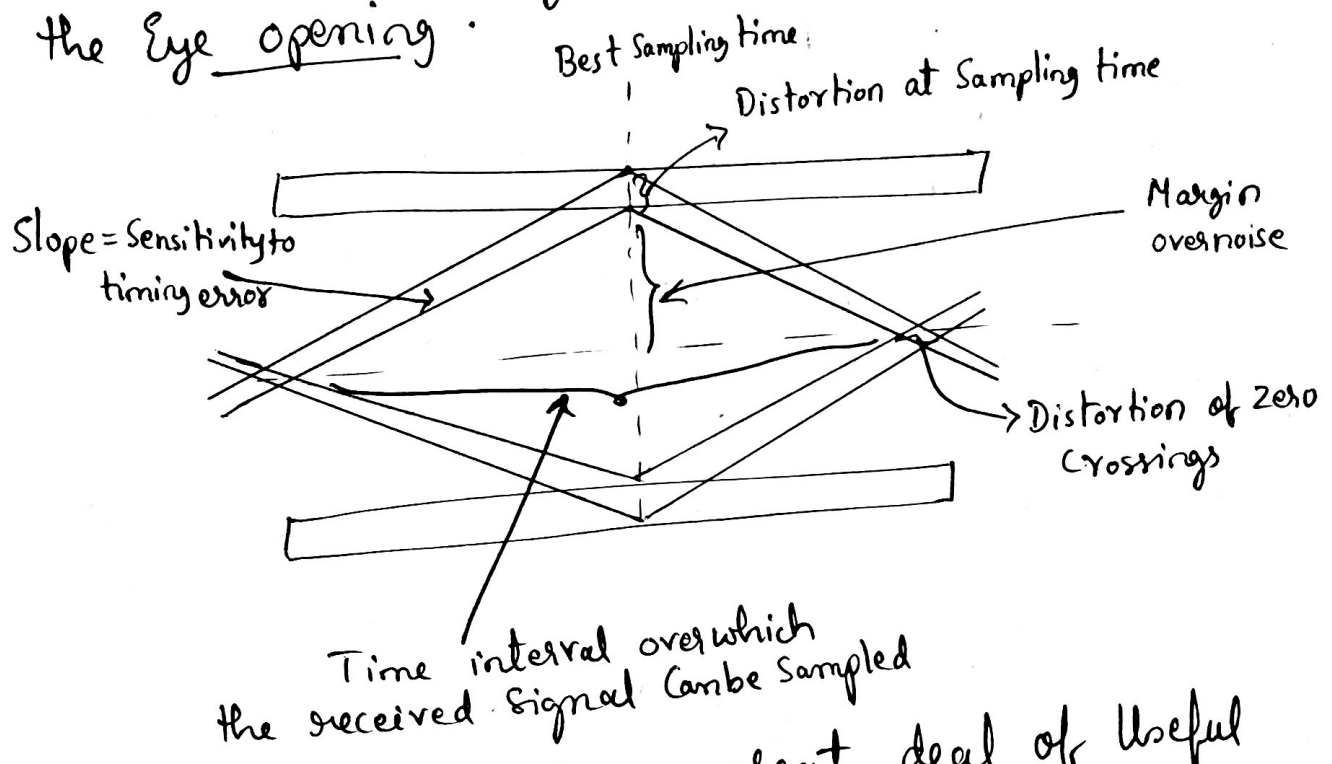
→ An operational tool for evaluating the effects of ISI in an insightful manner is the so-called eye pattern.

The eye pattern is defined as the Synchronized Superposition of all possible realizations of the signal.

interest (e.g., received signal, receiver output) Viewed within a particular signaling interval.

The eye pattern derives its name from the fact that it resembles the human-eye for binary waves.

The interior region of the eye pattern is called the Eye opening.



An eye pattern provides a great deal of useful information about the performance of a data transmission system.

- From eye pattern, we get
1. The width of the eye opening defines the time interval over which the received signal can be sampled without error from intersymbol interference.
 2. The sensitivity of the system to timing errors is determined by the rate of closure of the eye as the sampling time is varied.
 3. The height of the eye opening, at a specified sampling time, defines the Noise margin of the system.

*. NYQUIST CRITERION FOR DISTORTIONLESS TRANSMISSION

③. Explain the concept of Nyquist Criterion for distortionless transmission.

→ Consider the receive filter expression

$$y(t_i) = \mu a_i + \mu \sum_{\substack{k=-\infty \\ k \neq i}}^{\infty} a_k p[(i-k)T_b] + n(t_i) \quad - (1)$$

Sampling at $t = iT_b$.

where,
$$p(iT_b - kT_b) = \begin{cases} 1 & i=k \\ 0 & i \neq k \end{cases} \quad - (2)$$

$p(0) = 1$, by normalization
if $p(t)$ satisfies the condition of Equation (2), the receiver output $y(t_i)$ simplifies to

$$y(t_i) = \mu a_i \text{ for all } i$$

which implies zero ISI. Hence, the condition of Equation (2) ensures perfect reception in the absence of noise. This is in Time domain.

In frequency domain, we get

$$P_s(f) = R_b \sum_{n=-\infty}^{\infty} P(f - nR_b)$$

where

$$R_b = 2W \rightarrow \text{bit rate}$$

$$R_b = \frac{1}{T_b} \rightarrow \text{Nyquist rate}$$

and $W \rightarrow$ Nyquist bandwidth

i.e.,
$$\sum_{n=-\infty}^{\infty} P(f - nR_b) = T_b \quad - (3)$$

The Nyquist Criterion for distortionless baseband transmission in the absence of noise is stated as "The frequency function $P(f)$ eliminates ISI for samples taken at intervals

T_b provided that it satisfies equation (3)."

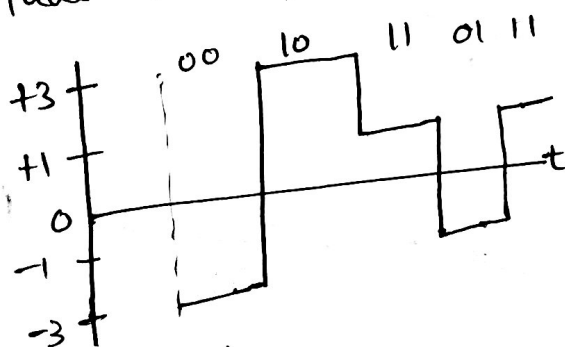
Note that $P(f)$ refers to overall system, incorporating the transmitter filter, the channel and the receive filter in accordance with

$$P(f) = G(f) \cdot H(f) \cdot C(f)$$

* BASEBAND M-ARY PAM TRANSMISSION

(4). Explain the concept of ~~Baseband~~ Baseband M-ARY PAM Transmission. (6 marks)

→ In a baseband M-ary PAM system, the pulse-amplitude Modulator produces one of M-possible amplitude levels, with $M > 2$, as shown.



(a) waveform

Dibit	Amplitude
00	-3
01	-1
11	+1
10	+3

(b) Representation of the four possible pairs of bits.

In an M-ary system, the information source emits a sequence of symbols from an alphabet that consists of M symbols. Each amplitude level at the PAM output corresponds to a distinct symbol, so that there are M-distinct amplitude levels to be transmitted.

In an M-ary PAM system, one baud (symbols per second) is equal to $\log_2 M$ bits per second.

∴ Symbol duration T is given by

$$T = T_b \cdot \log_2 M = T_b \cdot \frac{\log M}{\log 2}$$

⑤. Explain the bandwidth requirements of T_1 system. (6 marks)

→ The signal format for the T_1 carrier system is used to multiplex 24 independent voice inputs, based on an 8-bit PCM word.

The bit duration of the resulting TDM signal is $T_b = 0.647 \mu s$.

Assuming the use of an ideal Nyquist channel, the minimum transmission bandwidth B_T of the T_1 system is (for $\alpha=0$)

$$B_T = W = \frac{1}{2T_b} = 772 \text{ kHz}.$$

for $\alpha=1$,

$$B_T = W(1+\alpha) = 2W = \frac{1}{T_b} = 1.544 \text{ MHz}$$

To accommodate an FDM system using SSB modulation to transmit 24 independent voice inputs, and assuming a bandwidth of 4 kHz for each voice input, the channel must provide the transmission bandwidth

$$B_T = 24 \times 4k = \underline{96 \text{ kHz}}.$$

5. Noise.

Noise is an electronic signal that is a mixture of many random frequencies at many amplitudes that gets added to an information signal as it is transmitted from one place to another.

Noise examples are, hiss sound in the speaker, Snow in TV screen, etc.

(*) Signal to Noise Ratio, External Noise, Internal Noise.

6. Define Signal to Noise ratio (SNR). Explain the different types of External and Internal Noise. (8 marks)

→ The signal-to-noise (S/N) ratio, also designated SNR, indicates the relative strengths of the signal and the noise in a communication system.

The S/N ratio is computed by using either voltage or power value:

$$\boxed{S/N = \frac{V_s}{V_n}}$$

or

$$\boxed{S/N = \frac{P_s}{P_n}}$$

where V_s = signal voltage
 V_n = noise voltage

P_s = signal power

P_n = noise power

S/N values can be converted to decibels as follows:

For Voltage: $dB = 20 \log S/N$

For power: $dB = 10 \log S/N$

External Noise:

It comes from sources over which we have little or no control — industrial, atmospheric, or space.

Noise shows up as a random ac voltage.

The amplitude varies over a wide range, as does the frequency. This is generally known as white noise.

(i) Industrial Noise:

It is produced by manufacturing equipments, such as automotive ignition systems, electric motors and generators.

(ii) Atmospheric Noise:

The electrical disturbances that occur naturally in the earth's atmosphere are another source of noise. Atmospheric noise is often referred to as static. Static usually comes from lightning, the electric discharges that occur between clouds or between the earth and clouds.

(iii) Extraterrestrial Noise:

Extraterrestrial noise, ~~is the~~ solar and cosmic, comes from sources in space. One of the primary sources is the sun, which radiates a wide range of signals in a broad noise spectrum.

Noise generated by stars outside our solar system is generally known as Cosmic noise.

Internal Noise:

Electronic components in a receiver such as resistors, diodes, and transistors are major sources of internal noise.

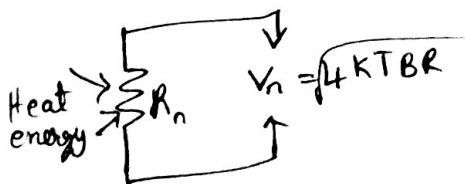
Most internal noise is caused by a phenomenon known as Thermal agitation, the random motion of free electrons in a conductor caused by heat. This is known as Thermal Noise.

The amount of open-circuit noise voltage appearing across a resistor or the input impedance to a receiver can be calculated according to Johnson's formula

$$V_n = \sqrt{4kTB R}$$

Where V_n = rms noise voltage
 k = Boltzmann's constant ($1.38 \times 10^{-23} \text{ J/K}$)
 T = temperature, K ($^{\circ}\text{C} + 273$)
 B = bandwidth, Hz
 R = resistance, Ω .

A resistor acts as a tiny generator of noise voltage.



Note: (1) Average noise power in Watts is given by

$$P_n = kTB$$

(2) $\sqrt{\text{rms}}$ noise current in a device

is given by

$$I_n = \sqrt{2qIB}$$

where q = Charge on an electron
 $q = 1.6 \times 10^{-19} \text{ C}$

I = direct current, A

B = Bandwidth, Hz.

* SEMICONDUCTOR NOISE:

Electronic components such as diodes and transistors are major contributors of noise. In addition to thermal noise, semiconductors produce shot noise, transit-time noise and flicker noise.

The most common type of semiconductor noise is shot noise.

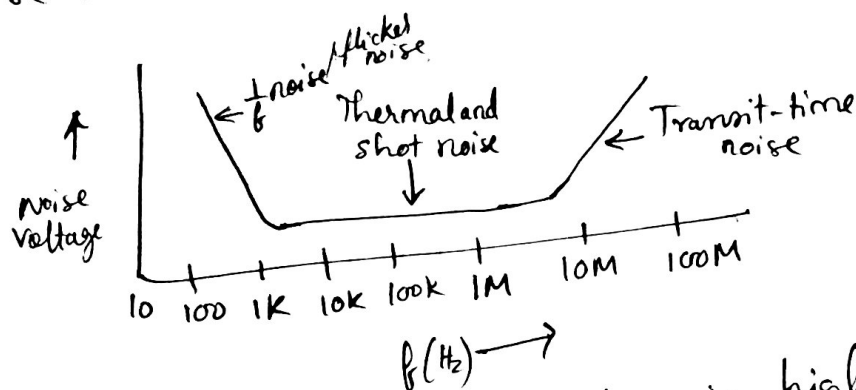
Shot noise is produced by the random movement of electrons or holes across a PN junction.

Another kind of noise that occurs in transistors is called transit-time noise. The term, transit time refers to how long it takes for a current carrier such as a hole or electron to move from the input to the output.

The third type of semiconductor noise is flicker noise or excess noise. occurs in resistors and conductors. This disturbance is the result of minute random variations of resistance in the semiconductor material.

It is directly proportional to current and temperature. and it is inversely proportional to frequency, and for this reason it is sometimes referred to as $\frac{1}{f}$ noise.

- Noise in a transistor with respect to frequency.



- At low frequencies, noise voltage is high, because of $\frac{1}{f}$ noise.
- Noise is lowest in the mid range, where most devices operate, due to thermal and shot effects.
- At very high frequencies, the rise in noise is due to transit-time effects near the upper cut-off frequency of the device.

* Expressing Noise Levels

The noise quality of a receiver can be expressed as in terms of noise figure, noise factor, noise temperature and SINAD.

- (i) Noise factor/noise ratio (NR): It is the ratio of the S/N power at the input to the S/N power at the output.

$$NR = \frac{S/N_{\text{input}}}{S/N_{\text{output}}}$$

(ii) Noise figure (NF):

When the noise factor is expressed in decibels, it is called the noise figure (NF)

$$NF = 10 \log NR \text{ dB.}$$

(iii) Noise temperature (T_N):

Noise temperature T_N is expressed in kelvins and is given by

$$T_N = 290(NR - 1)$$

(iv) SINAD:

SINAD is the Composite signal plus the noise and distortion, divided by noise and distortion contributed by the receiver. In symbolic form,

$$SINAD = \frac{S+N+D \text{ (composite signal)}}{N+D \text{ (receiver)}}$$

Distortion refers to the harmonics present in a signal caused by nonlinearities.

The SINAD is a power ratio, and it is almost always expressed in decibels:

$$SINAD = 10 \log \frac{S+N+D}{N+D} \text{ dB}$$

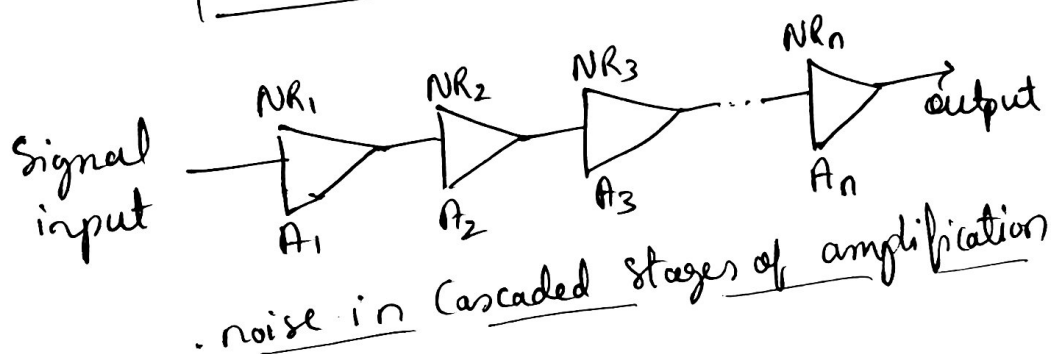
Noise in Cascaded stages

⑦ Illustrate the concept of noise in cascaded stages with a diagram. Write the Friis formula and mention its terms. (6 marks)

→ The formula used to calculate the overall noise performance of a receiver or multiple stages of RF amplification called Friis's formula.

Noise Ratio

$$NR = NR_1 + \frac{NR_2 - 1}{A_1} + \frac{NR_3 - 1}{A_1 A_2} + \dots + \frac{NR_n - 1}{A_1 A_2 \dots A_{n-1}}$$



Where

NR = noise ratio
 NR_1 = noise ratio of input or first amplifier
 NR_2 = noise ratio of second amplifier
 NR_3 = noise ratio of third amplifier & soon.
 A_2 = power gain of second amplifier
 A_1 = power gain of first amplifier
 A_3 = power gain of third amplifier & soon.

Noise figure is given by $NF = 10 \log NR$ dB

Problems:-

- ①. What is the open-circuit noise voltage across a $100\text{ k}\Omega$ resistor over the frequency range of direct current to 20 kHz at room temperature (25°C)?

$$\begin{aligned}\rightarrow V_n &= \sqrt{4kTBR} \\ &= \sqrt{4(1.38 \times 10^{-23})(25+273)(20 \times 10^3)(100 \times 10^3)} \\ V_n &= 5.74 \mu\text{V}\end{aligned}$$

- ②. The bandwidth of a receiver with a 75Ω input resistance is 6 MHz . The temperature is 29°C . What is the input thermal noise voltage?

$$\begin{aligned}\rightarrow T &= 29 + 273 = 302\text{ K} \\ V_n &= \sqrt{4kTBR} \\ &= \sqrt{4(1.38 \times 10^{-23})(302)(6 \times 10^6)(75)} \\ V_n &= 2.74 \mu\text{V}\end{aligned}$$

- ③. What is the average noise power of a device operating at a temperature of 90°F with a bandwidth of 30 kHz ?

$$\begin{aligned}\rightarrow T_C &= \frac{5(T_F - 32)}{9} = \frac{5(90 - 32)}{9} = \frac{5(58)}{9} = \frac{290}{9} = 32.2^\circ\text{C} \\ T_K &= T_C + 273 = 32.2 + 273 = 305.2\text{ K}\end{aligned}$$

$$\begin{aligned}\therefore \text{average noise power } P_n &= kTB \\ &= (1.38 \times 10^{-23})(305.2)(30 \times 10^3) \\ P_n &= 1.26 \times 10^{-16}\text{ W}\end{aligned}$$

④ An RF amplifier has an S/N ratio of 8 at the input and an S/N ratio of 6 at the output. What are the noise factor and noise figure?

→

$$NR = \frac{\text{S/N ratio at input}}{\text{S/N ratio at output}}$$

~~NR~~ $\frac{8}{6}$ noise factor

$$NR = \frac{8}{6} = 1.333$$

$$\begin{aligned} \therefore NF_{\text{noise figure}} &= 10 \log 1.333 \\ &= 10 (0.125) \\ &= 1.25 \text{ dB} \end{aligned}$$

⑤ If the noise ratio is 1.5 then what is the equivalent noise temperature

→

$$\begin{aligned} T_N &= 290 (NR - 1) \\ &= 290 (1.5 - 1) \\ &= 290 (0.5) \end{aligned}$$

$$T_N = 145 \text{ Kelvin}$$

⑥ A receiver with a 75Ω input resistance operates at a temperature of 31°C . The received signal is at 89 MHz with a bandwidth of 6 MHz . The received signal voltage of $8.3 \mu\text{V}$ is applied to an amplifier with a noise figure of 2.8 dB .

- Find
- (a) the input noise power
 - (b) the input signal power
 - (c) S/N, in decibels
 - (d) the noise factor & S/N of the amplifier
 - (e) noise temperature of the amplifier.

a. $T_K = 273 + 31 = 304 \text{ K}$

$$V_n = \sqrt{4 k T B R}$$

$$= \sqrt{4 (1.38 \times 10^{-23}) (304) (6 \times 10^6) (75)}$$

$$V_n = 2.75 \mu\text{V}$$

input noise power

$$P_n = \frac{(V_n)^2}{R} = \frac{(2.75 \times 10^{-6})^2}{75} = 0.1 \text{ pW}$$

b. $P_s = \frac{(V_s)^2}{R} = \frac{(8.3 \times 10^{-6})^2}{75} = 0.918 \text{ pW}$

c. $\frac{S}{N} = \frac{P_s}{P_n} = \frac{0.918}{0.1} = 9.18$

$$\text{dB} = 10 \log \frac{S}{N} = 10 \log 9.18 = 9.63 \text{ dB}$$

d.

$$\text{NF} = 10 \log \text{NR}$$

$$\text{NR} = \text{anti-log} \frac{\text{NF}}{10}$$

$$\text{NR} = 10^{\text{NF}/10}$$

$$\text{NR} = 10^{2.8/10} = 10^{0.28} = 1.9$$

$$\text{NR} = \frac{S/N \text{ input}}{S/N \text{ output}}$$

$$\therefore S/N \text{ output} = \frac{S/N \text{ input}}{\text{NR}} = \frac{9.18}{1.9} = 4.83$$

$$S/N \text{ (output)} = S/N \text{ (amplifier)} = 4.83$$

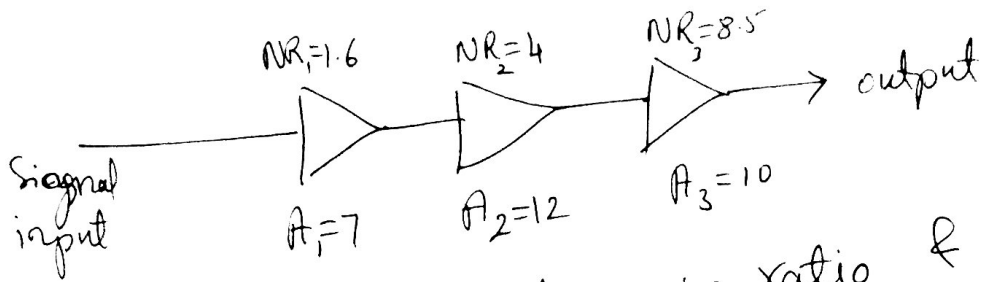
$$(e). T_N = 290(\text{NR} - 1)$$

$$= 290(1.9 - 1)$$

$$= 290(0.9)$$

$$T_N = 261 \text{ K}$$

Consider noise in cascaded stages of amplification as



Calculate the overall noise ratio & noise figure.

(Friis formula)

→

$$NR = NR_1 + \frac{NR_2 - 1}{A_1} + \frac{NR_3 - 1}{A_1 A_2}$$

$$= 1.6 + \frac{4 - 1}{7} + \frac{8.5 - 1}{(7)(12)}$$

$$= 1.6 + 0.4286 + 0.0893$$

$$NR = \underline{2.12}$$

$$NF = 10 \log NR$$

$$= 10 \log 2.12$$

$$= 10(0.326) = \underline{3.26 \text{ dB}}$$